Generalised Linear Models - Introduction

Why Generalised Linear Models (GLMs)?

Linear models suppose that we have observations

$$Y \sim N(X\beta, \sigma^2 I).$$

Can we use linear models for genetic case-control study?

Table: Data: Type 2 Diabetes example

Sample id	Y_i	X_i
1	type 2 diabetes	CC
2	no type 2 diabetes	CC
3	no type 2 diabetes	CT
4	no type 2 diabetes	TT
...	...	...
2000	type 2 diabetes	CC

Y_i : no type 2 diabetes $\rightarrow$ 0
type 2 diabetes $\rightarrow$ 1

X_i : CC $\rightarrow$ 0
CT $\rightarrow$ 1
TT $\rightarrow$ 2

GLMs allow for non-normal Y!

↳ Genotypes at a particular human genome location.

Generalised Linear Models - Outline

- Binomial regression
- Exponential families
- Generalised linear models (GLMs)
- Quasilikelihood and overdispersion

Binomial Regression I

Learning goals

Understand binomial regression

- know when you should use binormal regression.
- be able to write binomial regression model and its likelihood.
- be able to obtain estimators of parameters or function of parameters using R script.
- be able to quantify uncertainty of the estimators (e.g., computing CI).
- be able to test hypothesis.
- be able to do model selection.

Understand asymptotic properties of MLEs (maximum likelihood estimators)

- use asymptotic normality of MLEs to quantify uncertainty of the estimators (e.g., Wald CI).

Undertand Wald test and likelihood ratio test (LRT)

- use them to test hypothesis in binomial regression

Understand (scaled) deviance

- use it to test model adequacy or perform LRT and model comparison.

Challenger example

Challenger disaster (Challenger space shuttle)

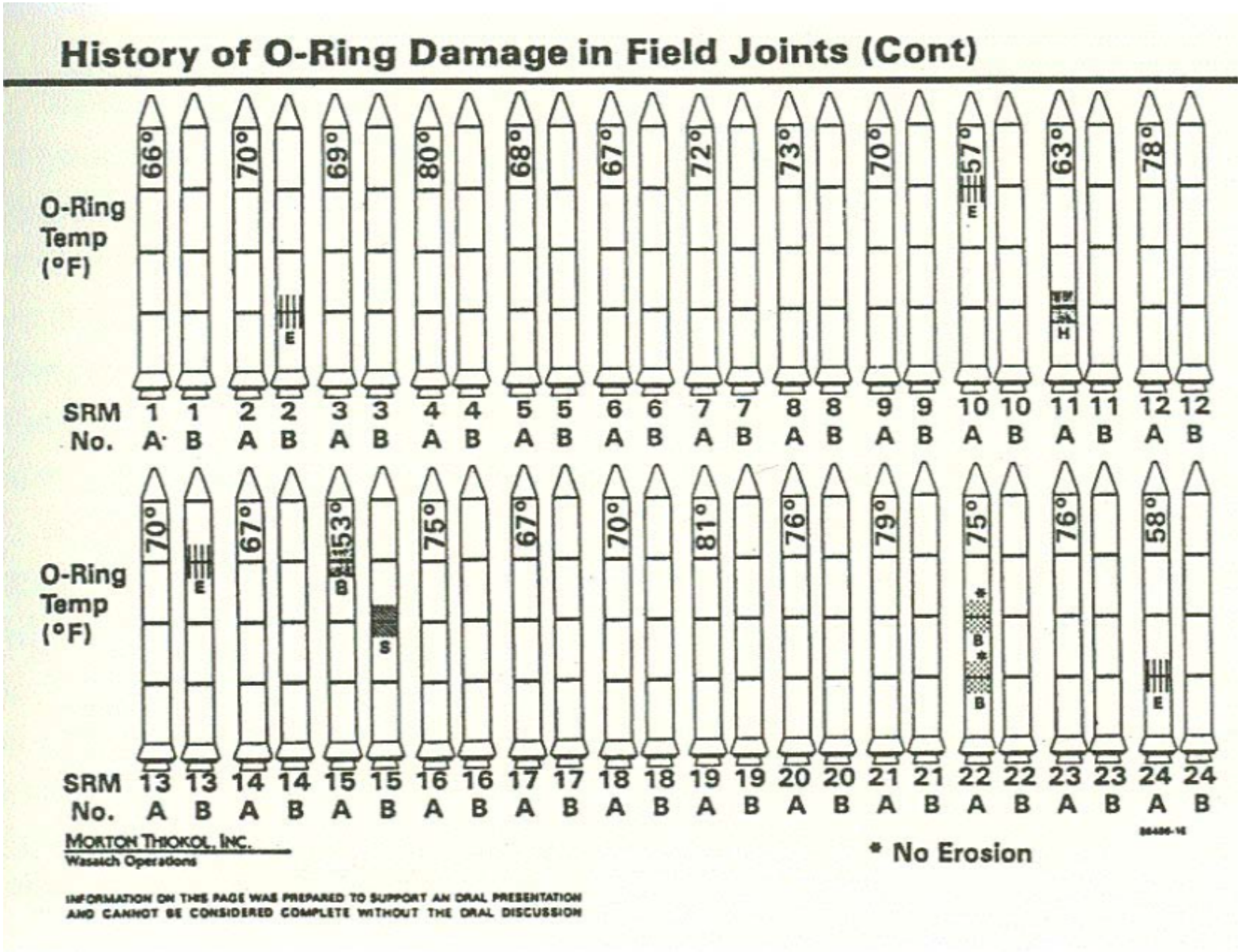
On the 28th of January 1986 the Space Shuttle Challenger broke apart after an (*) O-ring seal failed at liftoff, leading to the deaths of its seven crew members.

Despite concerns about the O-rings failing due to the cold—the forecast temperature was $29 \pm 3\,^oF$—no one was able to provide an analysis that convinced NASA (who were under pressure to launch following several previous delays) not to go ahead.

The way the data was presented didn't help matters.

(*) O-rings → Large rubber seals in the joints between sections of the Challenger's solid rocket boosters → To prevent hot high pressure gases escaping through the booster joints during launch.

Challenger disaster: data



Challenger disaster: data

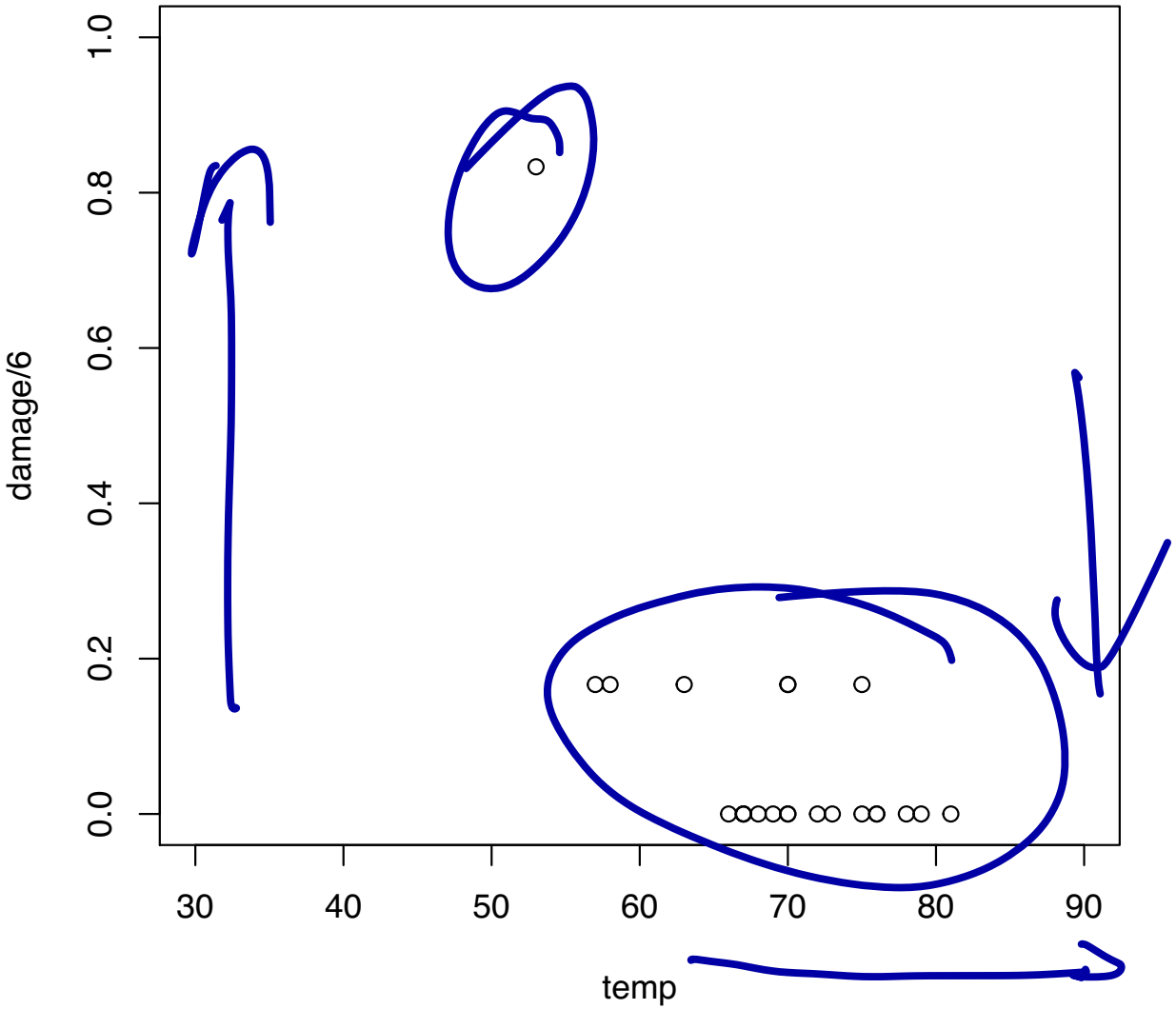
Summarise data using number of damaged O-rings (out of 6) per launch.

Data is in dataframe `orings`.
Response `damage` is number of damaged O-rings (out of 6).
Predictor `temp` is temperature (^oF)

```
> library(faraway)
> data(orings)
> str(orings)
data.frame: 23 obs. of  2 variables:
 $ temp  : num  53 57 58 63 66 67 67 67 68 69 ...
 $ damage: num  5 1 1 1 0 0 0 0 0 0 ...
> plot(damage/6 ~ temp, data = orings,
+      xlim = c(30, 90), ylim = c(0, 1))
```

dim(orings)

Challenger disaster: visualize data



Challenger disaster: model

Make the assumption that Y_i, the number of damaged O-rings on the i-th launch, has distribution

$$Y_i \sim \text{bin}(6, p_i)$$

where p_i depends on the temperature t_i. We also assume that the Y_i are independent.

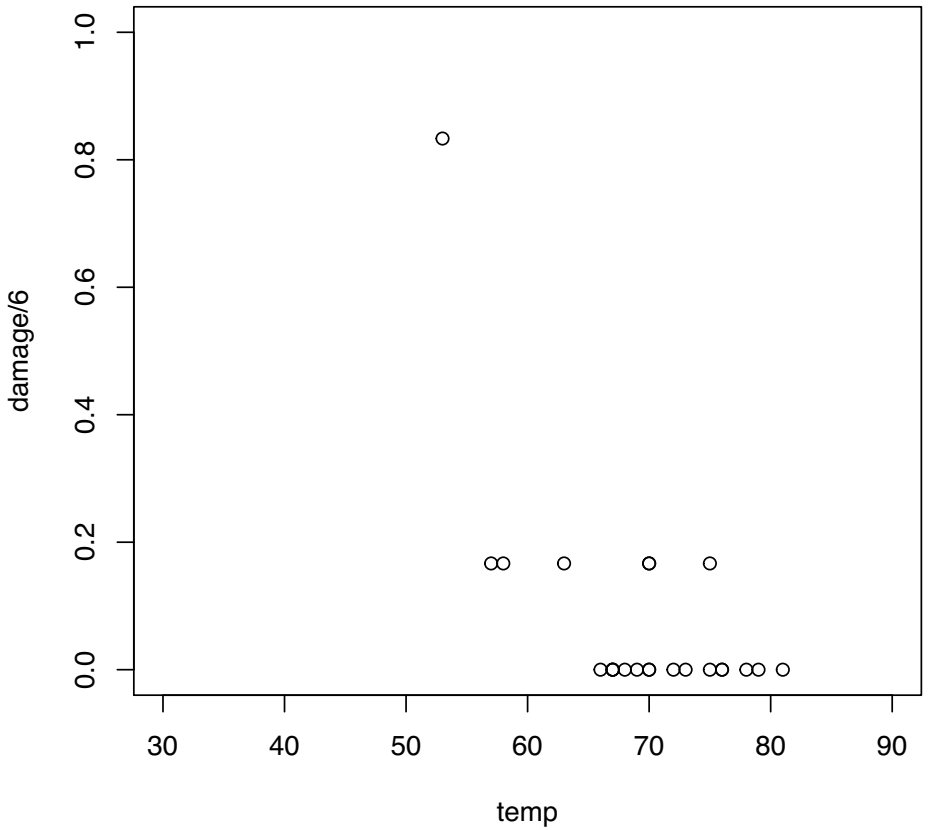
How to relate p_i with t_i.

- Question: just use a linear relationship? $p_i = \beta_0 + \beta_1 t_i$?

No. We need parameter bounds so $0 \leq p_i \leq 1$. We need alternate function to relate p_i with t_i.

Alternative function to relate p_i with t_i

For a single launch, best estimate of p_i is just $y_i/6$. From plot of $y_i/6$ against t_i it is reasonable to assume that $p_i = p(t_i)$ where p is a smooth function of the temperature, decreasing from 1 down to 0 as the temperature increases.



We choose logistic function: suppose that for some β_0 and β_1

$$p(t) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 t)}} = \frac{e^{\beta_0 + \beta_1 t}}{1 + e^{\beta_0 + \beta_1 t}}$$

- Restricts $0 \leq p(t) \leq 1$
- Monotonic increasing/decreasing function
- Can be linearized through the *logit* transformation

$$\log \frac{p(t)}{1 - p(t)} = \beta_0 + \beta_1 t$$

$p \in (0,1)$

$\frac{p}{1-p} \in (0, \infty)$

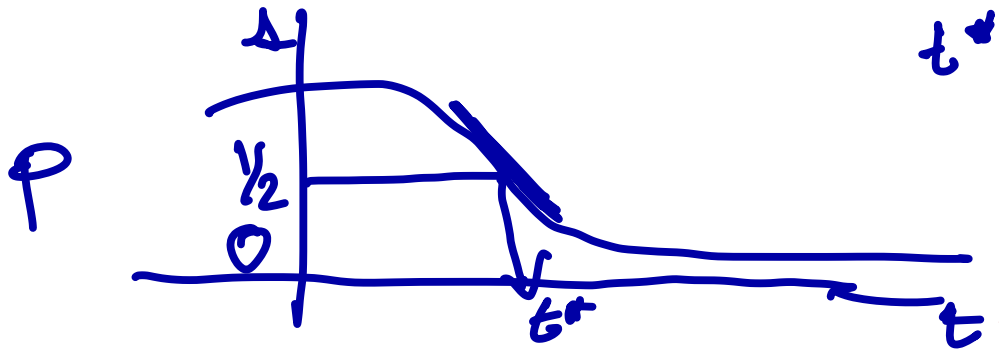
$\log\left(\frac{p}{1-p}\right) \in (-\infty, \infty)$

We choose logistic function: suppose that for some β_0 and β_1

$$p(t) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 t)}} = \frac{e^{\beta_0 + \beta_1 t}}{1 + e^{\beta_0 + \beta_1 t}}$$

- $p(t) = 1/2$ for $t = -\beta_0/\beta_1$, so $-\beta_0/\beta_1$ controls the location of the curve.
- $p'(-\beta_0/\beta_1) = \beta_1/4$, so β_1 controls the steepness of the curve.

Example: see R script and result in the section "logistic function with different values for beta0 and beta1" of Challenger.pdf.



Challenger disaster: model

Make the assumption that Y_i, the number of damaged O-rings on the i-th launch, has distribution

$$Y_i \sim \text{bin}(6, p_i)$$

where

$$p_i = \frac{e^{\beta_0 + \beta_1 t_i}}{1 + e^{\beta_0 + \beta_1 t_i}}.$$

We also assume that the Y_i are independent.

Challenger disaster: model fitting

Make the assumption that Y_i, the number of damaged O-rings on the i-th launch, has distribution

$$Y_i \sim \text{bin}(6, p_i)$$

where

$$p_i = \frac{e^{\beta_0 + \beta_1 t_i}}{1 + e^{\beta_0 + \beta_1 t_i}}.$$

We also assume that the Y_i are independent.

Maximum likelihood estimators (MLE) $\hat{\beta}_0$, $\hat{\beta}_1$: values for β_0, β_1 which maximize the log-likelihood.

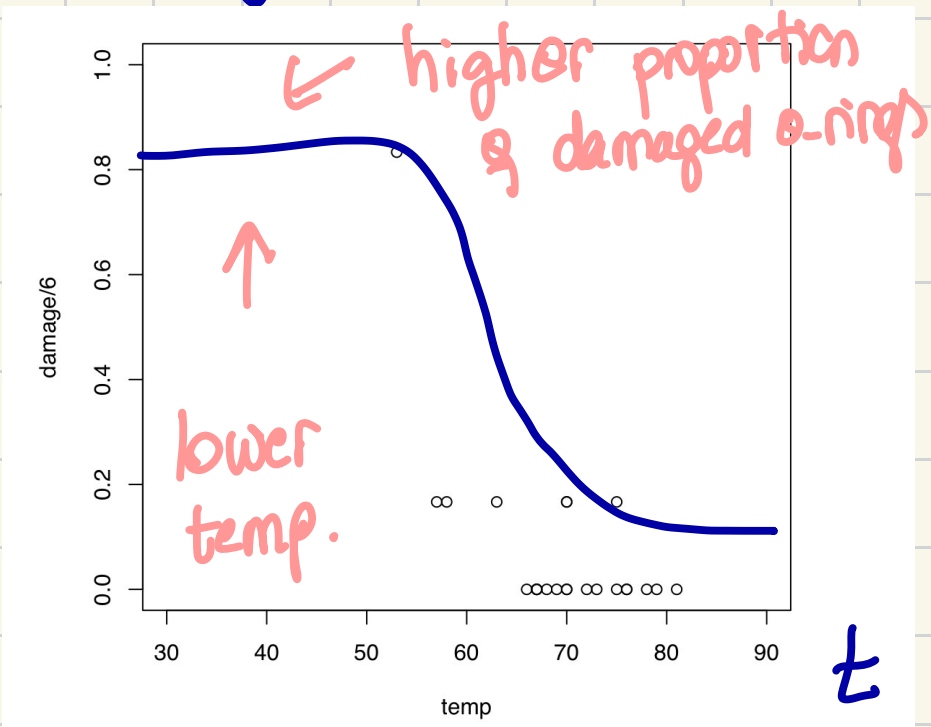
Challenger disaster: model fitting

The log-likelihood is

$$
\begin{aligned}
l(\beta_0, \beta_1) &= \log \mathcal{L}(\beta_0, \beta_1) \\
&= \log \mathbb{P}(\mathrm{Y} = \mathrm{y} \mid \beta_0, \beta_1) = \log \prod_i \mathbb{P}(Y_i = y_i \mid \beta_0, \beta_1) \\
&= \sum_i \log \left(\binom{6}{y_i} p_i^{y_i} (1 - p_i)^{6 - y_i} \right) \\
&= c + \sum_i (y_i \log p_i + (6 - y_i) \log(1 - p_i)) \\
&= c + \sum_i \left(y_i \log \frac{p_i}{1 - p_i} + 6 \log(1 - p_i) \right)
\end{aligned}
$$

Put $\eta_i = \beta_0 + \beta_1 t_i$, then $\log p_i/(1 - p_i) = \eta_i$ and $\log(1 - p_i) = -\log(1 + e^{\eta_i})$.

Thursday recap → Binomial regression



← higher proportion of damaged o-rings

↑ lower temp.

t

$p = \frac{Y}{6}$

?

$\frac{Y}{6} = p(t)$

$$p(t) = \frac{1}{1+e^{-(\beta_0+\beta_1 t)}} = \frac{e^{\beta_0+\beta_1 t}}{1+e^{\beta_0+\beta_1 t}}$$

$\in (0,1)$

- Why this p(t) ?

$$\log\left(\frac{p(t)}{1-p(t)}\right) = \beta_0 + \beta_1 t$$

$\in \mathbb{R}$

logit transformation

"2" linear predictor

Goal → Inference about β_0 and β_1

Approach → MLE → Maximizing LL

How ?

$$Y_i \sim Bin(n, p_i)$$

The log-likelihood is → where are β_0 & β_1 ?!

$$\begin{aligned}
l(\beta_0, \beta_1) &= \log \mathcal{L}(\beta_0, \beta_1) \\
&= \log \mathbb{P}(\mathbf{Y} = \mathbf{y} \mid \beta_0, \beta_1) = \log \prod_i \mathbb{P}(Y_i = y_i \mid \beta_0, \beta_1) \\
&= \sum_i \log \left(\binom{6}{y_i} p_i^{y_i} (1 - p_i)^{6 - y_i} \right) \\
&= c + \sum_i (y_i \log p_i + (6 - y_i) \log(1 - p_i)) \\
&= c + \sum_i \left(y_i \log \frac{p_i}{1 - p_i} + 6 \log(1 - p_i) \right)
\end{aligned}$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i = \eta_i \quad \Rightarrow \log(1-p_i) = -\log(1+e^{\eta_i})$$

↳ linear predictor

Challenger disaster: model fitting

There is no closed form solution for MLE $\hat{\beta}_0$, $\hat{\beta}_1$ in this model. Numerical search procedures are required to find MLE.

- the `glm` function uses the iterative weighted least squares (IWLS) algorithm - I will cover this later.
- For now, let's use the `optim` function.

Example:

- See R script and result in "maximum likelihood fitting" of Challenger.pdf
- $\hat{\beta}_0 = 11.667$ and $\hat{\beta}_1 = -0.216$

Challenger disaster

Heejung Shim

install.package faraway

```
#install.packages("faraway")
```

→ Only one time!

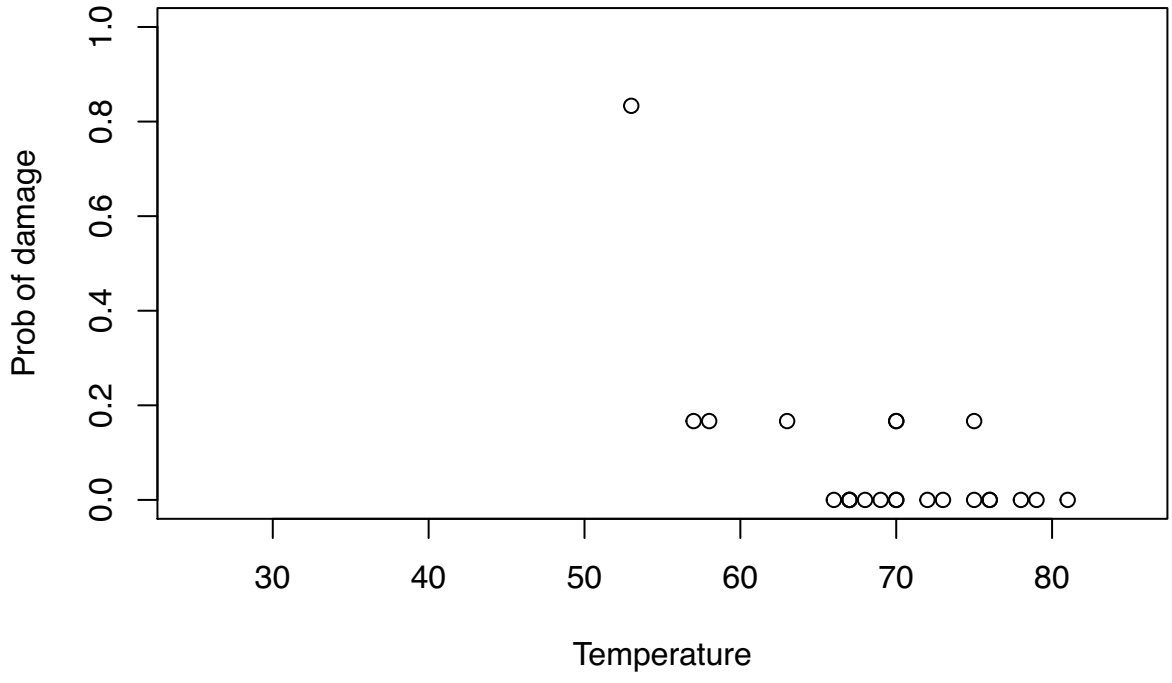
load and plot data

```
library(faraway)
data(orings)
str(orings)
```

→ Load the library in each R session

```
## 'data.frame':    23 obs. of  2 variables:
##  $ temp  : num  53 57 58 63 66 67 67 67 68 69 ...
##  $ damage: num  5 1 1 1 0 0 0 0 0 0 ...
```

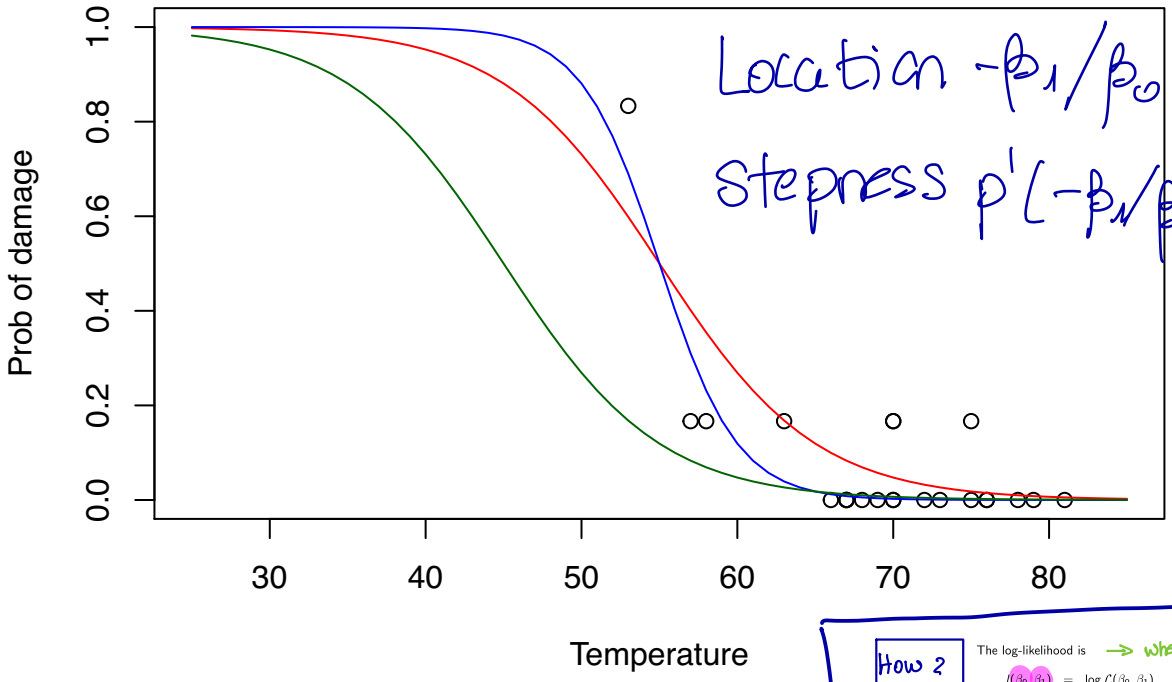
```
plot(damage/6 ~ temp, orings, xlim=c(25,85), ylim=c(0,1),
     xlab="Temperature", ylab="Prob of damage")
```



logistic function with different values for beta0 and beta1

$$p(t) = \frac{1}{1+e^{-(\beta_0+\beta_1 t)}} = \frac{e^{\beta_0+\beta_1 t}}{1+e^{\beta_0+\beta_1 t}}$$

```
try <- function(a, b, col = "red") {
  t <- seq(25, 85, 1)
  p <- 1/(1 + exp(-a - b*t))
  lines(t, p, col = col)
}
plot(damage/6 ~ temp, orings, xlim=c(25,85), ylim=c(0,1),
     xlab="Temperature", ylab="Prob of damage")
try(11, -0.2, col="red")
## Compared to red curve: same location, stronger steepness
try(22, -0.4, col="blue")
## Compared to red curve: shifted location, same steepness
try(9, -0.2, col="darkgreen")
```



Location $-\beta_1/\beta_0$

Steepness $p'(-\beta_1/\beta_0) = \frac{\beta_1}{4}$

maximum likelihood fitting

Define the log likelihood

```
logL <- function(beta, orings) {
  eta <- cbind(1, orings$temp) %*% beta
  return( sum( orings$damage*eta - 6*log(1 + exp(eta)) ) )
}
```

Find MLE using optim function

```
(betahat <- optim(c(10, -.1), logL, orings=orings, control=list(fnscale=-1))$par)
```

```
## [1] 11.6671414 -0.2162982
```

plot fitted model

```
plot(damage/6 ~ temp, orings, xlim=c(25,85), ylim=c(0,1),
     xlab="Temperature", ylab="Prob of damage")
```

Optim → Greedy algorithm

- Starting values are very important
- How can we decide on the starting values?



Solution:

θ_0^1 — θ_0^n ← possible started values

$f(\theta_0^1)$ — $f(\theta_0^n)$

↳ find the max

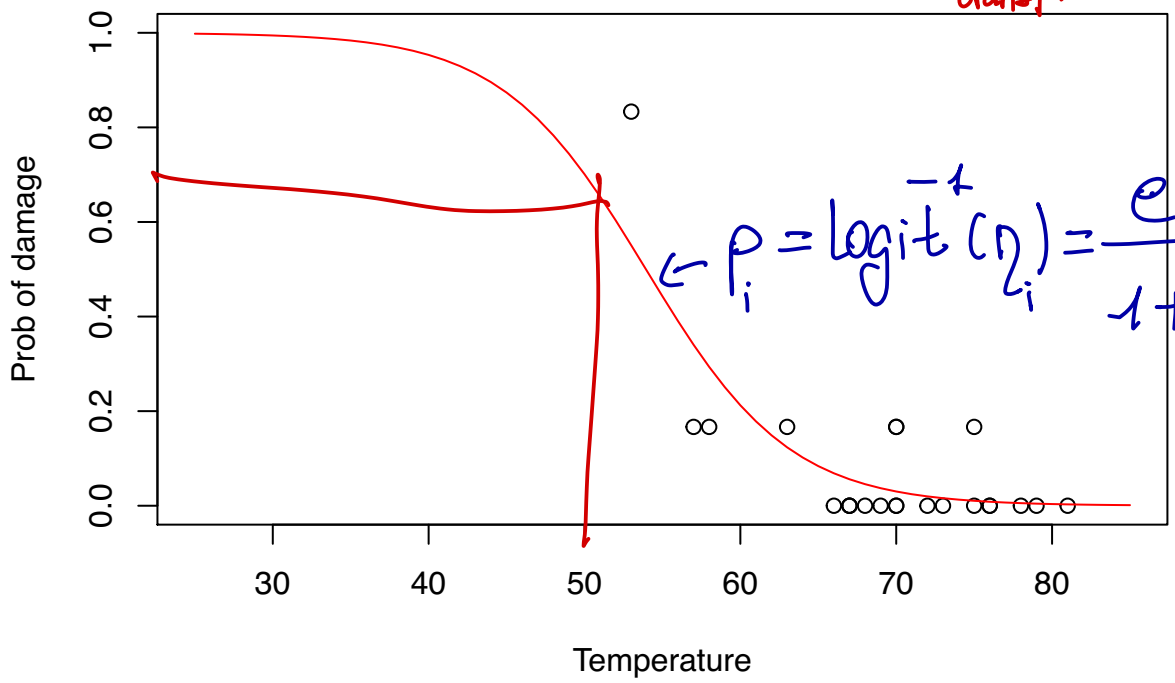
Logit transformation

$$\log\left(\frac{p_i}{1-p_i}\right) = \text{logit}(p_i) = \eta_i \Rightarrow \text{logit}^{-1}(\eta_i) = p_i$$

$\eta_i = \beta_0 + \beta_1 t_i$

logistic transf.

```
x <- seq(25,85,1)
ilogit <- function(x) exp(x)/(1+exp(x))
lines(x, ilogit(betahat[1] + betahat[2]*x), col="red")
```



p

$$p_i = \text{logit}^{-1}(\eta_i) = \frac{e^{\beta_0+\beta_1 t_i}}{1+e^{\beta_0+\beta_1 t_i}}$$

prediction for temp of 29

```
ilogit (betahat[1] + betahat[2]*29)
```

$$\text{logit}^{-1}(\hat{\eta}_i) = \text{logit}^{-1}(\hat{\beta}_0+\hat{\beta}_1 t_i)$$

$$\hat{p}_i !$$

```
## [1] 0.995479
```

Using the glm command instead

```
logitmod <- glm(cbind(damage,6-damage) ~ temp, family=binomial, orings)
summary(logitmod)
```

```
##
## Call:
## glm(formula = cbind(damage, 6 - damage) ~ temp, family = binomial,
##     data = orings)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -0.9529  -0.7345  -0.4393  -0.2079   1.9565
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) 11.66299    3.29626   3.538 0.000403 ***
## temp        -0.21623    0.05318  -4.066 4.78e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
##     Null deviance: 38.898  on 22  degrees of freedom
## Residual deviance: 16.912  on 21  degrees of freedom
## AIC: 33.675
##
## Number of Fisher Scoring iterations: 6
```

```
predict(logitmod, newdata=list(temp=29), type="response")
```

```
##         1
## 0.9954687
```

Confidence Interval for p

Compute standard errors

```
phat <- ilogit(betahat[1] + orings$temp*betahat[2])
I11 <- sum(6*phat*(1 - phat))
I12 <- sum(6*orings$temp*phat*(1 - phat))
I22 <- sum(6*orings$temp^2*phat*(1 - phat))
Iinv <- solve(matrix(c(I11, I12, I12, I22), 2, 2))
sqrt(Iinv[1,1])
```

```
## [1] 3.296634
```

```
sqrt(Iinv[2,2])
```

```
## [1] 0.05318407
```

Compute CI for eta = beta0 + beta1 29

```
si2 <- matrix(c(1, 29), 1, 2) %*% Iinv %*% matrix(c(1, 29), 2, 1)
etahat = betahat[1] + betahat[2]*29
eta_l = etahat - 2*sqrt(si2)
eta_r = etahat + 2*sqrt(si2)
etahat
```

```
## [1] 5.394493
```

```
c(eta_l, eta_r)
```

```
## [1] 1.851533 8.937452
```

Compute CI for p

```
ilogit (etahat)
```

```
## [1] 0.995479
```

```
c(ilogit (eta_l), ilogit (eta_r))
```

```
## [1] 0.8643070 0.9998686
```

Wald Test

Compute MLE

```
library(faraway)
data(orings)
logL <- function(beta, orings) {
```

Challenger disaster: questions

- Forecast probability of an O-ring being damaged when the launch temperature is 29 ^oF.
- How good is our forecast? Can we provide a confidence interval?
- Is temperature useful to predict the O-ring failing?

↳ Let's first formalize how binomial regression works

Binomial regression

Binomial regression model

We suppose that we observe $Y_i \sim \text{bin}(m_i, p_i)$, $i = 1, \ldots, n$, independent.

Goal: Connect/link p_i with some predictors x_i

The m_i are known and we suppose that for some **link function** g,

$$g(p_i) = \eta_i = \mathrm{x}_i^T \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_q x_{iq}$$

where x_i are known predictors and $\boldsymbol{\beta}$ are unknown parameters.

$\eta_i \rightarrow$ linear predictor

$g \rightarrow$ Link function $\rightarrow$ Challenger example $g \equiv$ logit

Binomial regression model: link function

Usual choices for g:

logit

$$\eta = \log \frac{p}{1-p}, \quad p = \frac{1}{1+\exp(-\eta)} = \frac{\exp(\eta)}{1+\exp(\eta)}$$

complementary log-log

$$\eta = \log(-\log(1-p)), \quad p = 1 - \exp(-e^{\eta})$$

probit

$$\eta = \Phi^{-1}(p), \quad p = \Phi(\eta),$$

where Φ is the cumulative distribution function (cdf) of the standard normal distribution.

Binomial regression model: link function

$$\eta = \log \frac{p}{1-p}, \quad p = \frac{1}{1+\exp(-\eta)} = \frac{\exp(\eta)}{1+\exp(\eta)}$$

logistic

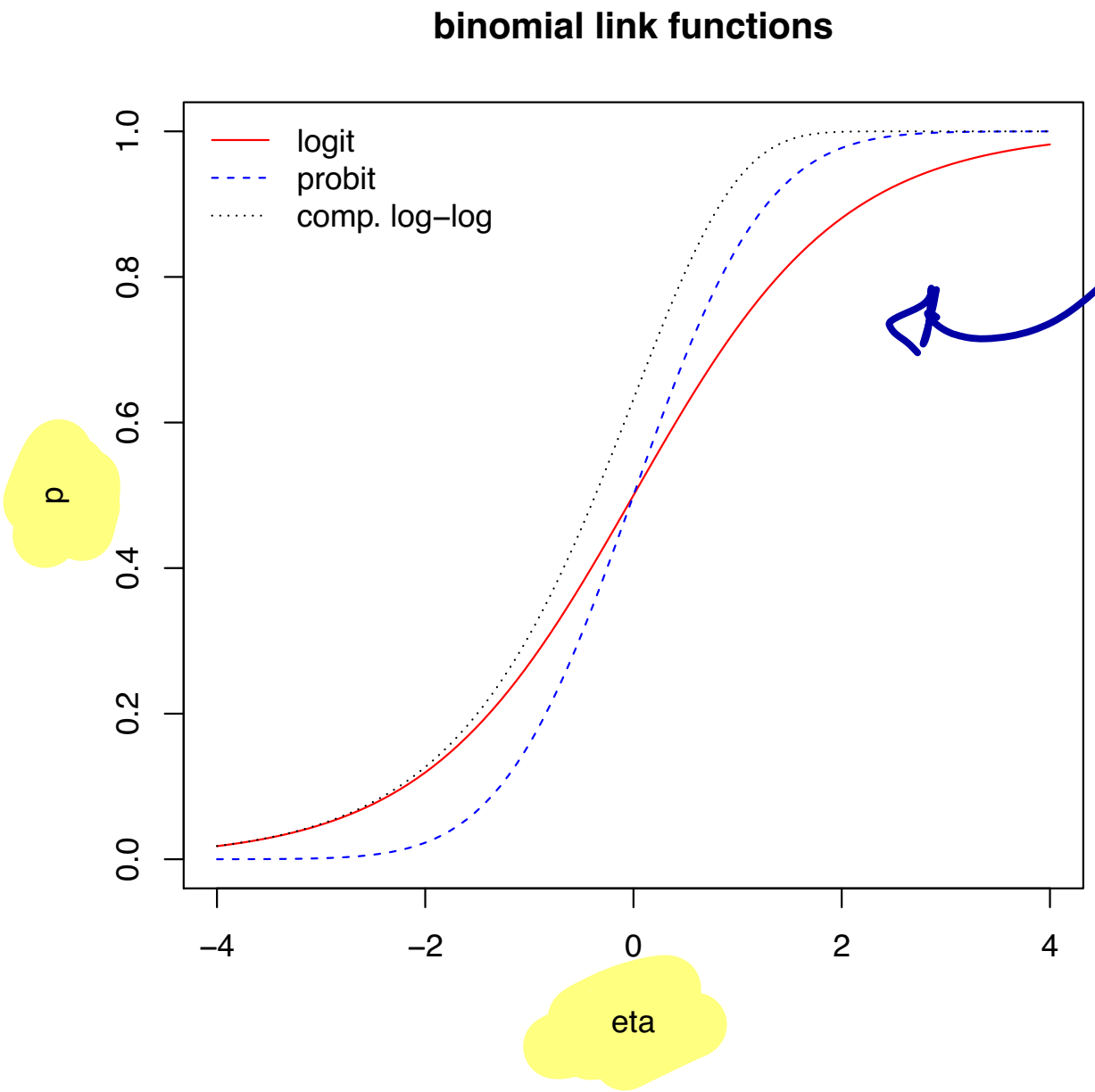
probit

log log

```
> curve(1/(1+exp(-x)), -4, 4, ylim=c(0,1),
+       xlab="eta", ylab="p", col="red",
+       main="binomial link functions")
> curve(pnorm(x), -4, 4, add=TRUE, col="blue", lty=2)
> curve(1-exp(-exp(x)), -4, 4, add=TRUE, col="black", lty=3)
> legend("topleft", c("logit", "probit", "comp. log-log"),
+        col=c("red", "blue", "black"), lty=c(1,2,3), bty="n")
```

↳ inverse of the link functions

Binomial regression model: link function



Inverse of the link functions

Model selection:

AIC, BIC, ...

Binomial regression model: likelihood

Given observations y_i of $Y_i \sim \text{bin}(m_i, p_i = g^{-1}(\eta_i))$, where $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$, the log-likelihood is

Linear combination of the predictors/parameters

$$
\begin{aligned}
l(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \log \mathbb{P}(Y_i = y_i) \\
&= \sum_{i=1}^{n} \log \left(\binom{m_i}{y_i} p_i^{y_i} (1-p_i)^{m_i - y_i} \right) \\
&= c + \sum_{i=1}^{n} y_i \log(g^{-1}(\eta_i)) + (m_i - y_i) \log(1 - g^{-1}(\eta_i))
\end{aligned}
$$

There is no closed form solution for MLE of $\boldsymbol{\beta}$. Numerical search procedures are required to find MLE.

Reminder: questions from Challenger disaster

- Forecast probability of an O-ring being damaged when the launch temperature is $29\,^oF$.
- How good is our forecast? Can we provide a confidence interval?
- Is temperature useful to predict the O-ring failing?

Forecast probability of an O-ring being damaged when the launch temperature is 29 ^oF.

$g(p) = \eta = \beta_0 + \beta_1 t \rightarrow \hat{p} = g^{-1}(\hat{\eta})$ for $t = 29$

If $g(p) = \log\left(\frac{p}{1-p}\right)$

$\hat{p} = g^{-1}(\hat{\eta})$ when $t = 29$.

$\Rightarrow \boxed{\hat{p} = \frac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}}}$

$\hat{p} = \frac{\exp(\hat{\eta})}{1+\exp(\hat{\eta})}$, where $\hat{\eta} = \hat{\beta}_0 + \hat{\beta}_1 29$

$\hat{p} = 0.995$: see R script and result in "prediction for temp of 29" of Challenger.pdf

$\hat{\beta}_0, \hat{\beta}_1 \rightarrow$ MLE $<$ optim / glm

Damage probability

$\hat{p} = 0.995$

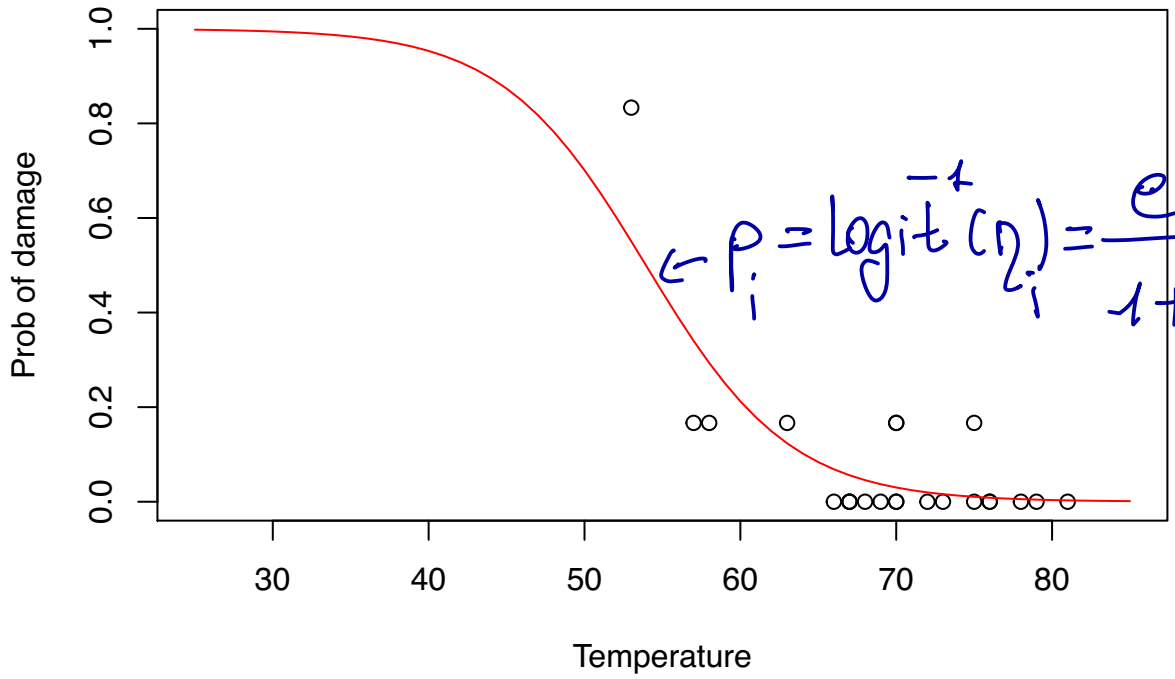
it is very high!

logistic transformation

$$\log\left(\frac{p_i}{1-p_i}\right) = \text{logit}(p_i) = \eta_i \Rightarrow \text{logit}^{-1}(\eta_i) = p_i$$

Logistic function $= \text{logit}^{-1}$ $\quad \eta_i = \beta_0 + \beta_1 t_i$

```
x <- seq(25,85,1)
ilogit <- function(x) exp(x)/(1+exp(x))
lines(x, ilogit(betahat[1] + betahat[2]*x), col="red")
```



p

$$p_i = \text{logit}^{-1}(\eta_i) = \frac{e^{\beta_0+\beta_1 t_i}}{1+e^{\beta_0+\beta_1 t_i}}$$

prediction for temp of 29

```
ilogit (betahat[1] + betahat[2]*29)
```

$$\text{logit}^{-1}(\hat{\eta}_i) = \text{logit}^{-1}(\hat{\beta}_0+\hat{\beta}_1 t_i) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 t_i}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 t_i}}$$

```
## [1] 0.995479
```

Using the glm command instead

```
logitmod <- glm(cbind(damage,6-damage) ~ temp, family=binomial, orings)
summary(logitmod)
```

```
##
## Call:
## glm(formula = cbind(damage, 6 - damage) ~ temp, family = binomial,
##     data = orings)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -0.9529  -0.7345  -0.4393  -0.2079   1.9565
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) 11.66299    3.29626   3.538 0.000403 ***
## temp        -0.21623    0.05318  -4.066 4.78e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

We will learn all about these

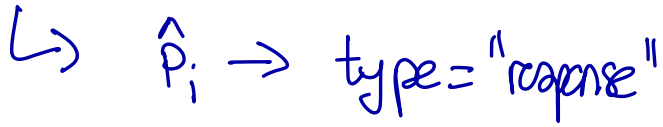
$$Y \sim Bin(n, p)$$

$$\begin{bmatrix} y_1 & n-y_1 \\ | & \\ y_{23} & n-y_{23} \end{bmatrix}$$

```
##
##     Null deviance: 38.898  on 22  degrees of freedom
## Residual deviance: 16.912  on 21  degrees of freedom
## AIC: 33.675
##
## Number of Fisher Scoring iterations: 6
```

```
predict(logitmod, newdata=list(temp=29), type="response")
```

```
##         1
## 0.9954687
```



Confidence Interval for p

Compute standard errors

```
phat <- ilogit(betahat[1] + orings$temp*betahat[2])
I11 <- sum(6*phat*(1 - phat))
I12 <- sum(6*orings$temp*phat*(1 - phat))
I22 <- sum(6*orings$temp^2*phat*(1 - phat))
Iinv <- solve(matrix(c(I11, I12, I12, I22), 2, 2))
sqrt(Iinv[1,1])
```

```
## [1] 3.296634
```

```
sqrt(Iinv[2,2])
```

```
## [1] 0.05318407
```

Compute CI for eta = beta0 + beta1 29

```
si2 <- matrix(c(1, 29), 1, 2) %*% Iinv %*% matrix(c(1, 29), 2, 1)
etahat = betahat[1] + betahat[2]*29
eta_l = etahat - 2*sqrt(si2)
eta_r = etahat + 2*sqrt(si2)
etahat
```

```
## [1] 5.394493
```

```
c(eta_l, eta_r)
```

```
## [1] 1.851533 8.937452
```

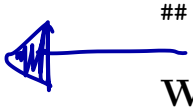
Compute CI for p

```
ilogit (etahat)
```

```
## [1] 0.995479
```

```
c(ilogit (eta_l), ilogit (eta_r))
```

```
## [1] 0.8643070 0.9998686
```



Wald Test

Compute MLE

```
library(faraway)
data(orings)
logL <- function(beta, orings) {
```

Reminder: questions from Challenger disaster

- Forecast probability of an O-ring being damaged when the launch temperature is 29 ^oF.
- How good is our forecast? Can we provide a confidence interval?
- Is temperature useful to predict the O-ring failing?

We need to know properties of MLE!!

Linear Models

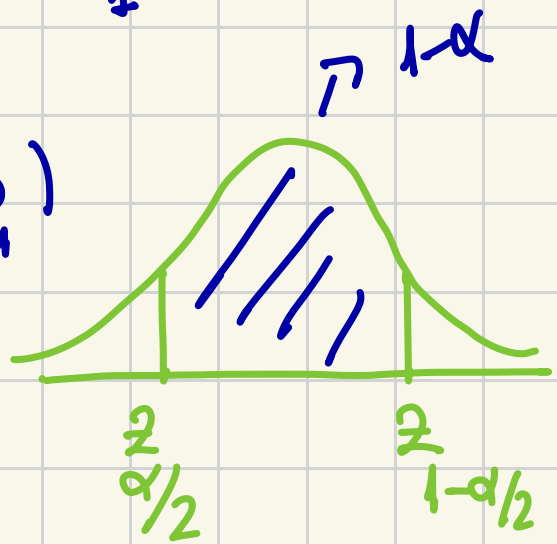
known

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad , \quad \varepsilon_i \sim N(0, \sigma^2)$$

Least squares $\longrightarrow \hat{\beta}_0 , \hat{\beta}_1$

Distribution $\hat{\beta}_1 \sim N(\beta_1, se^2(\hat{\beta}_1))$

α-level CI $\rightarrow \hat{\beta}_1 \pm z_{1-\alpha/2} \, se(\hat{\beta}_1)$
(i.e 95%)



Asymptotic properties MLE

Reminder: Maximum likelihood estimate (MLE)

Suppose that Y_i, $i = 1, \ldots, n$, are indepedendent, with densities/mass-functions $f_i(\cdot; \boldsymbol{\theta})$.

Given observations y_i of the Y_i, the log-likelihood is

$$l(\boldsymbol{\theta}) = l(\boldsymbol{\theta}; \mathrm{y}) = \sum_i \log f_i(y_i; \boldsymbol{\theta}).$$

f depends on some covariates

MLE $\hat{\boldsymbol{\theta}}$ is that value of $\boldsymbol{\theta}$ which maximises $l(\boldsymbol{\theta})$.

Note that allowing f_i to depend on i means that we can include the case where the distribution of Y_i depends on some covariate x_i. That is, we can have $f_i(\cdot; \boldsymbol{\theta}) = f(\cdot; \mathrm{x}_i, \boldsymbol{\theta})$ for some common f.

In linear models $Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$

Asymptotic properties of MLE

$n \to \infty$

Under certain regularity conditions (to be introduced later), the MLE is

- asymptotically consistent,
- asymptotically normal,
- asymptotically efficient.

MLE: asymptotic consistency

Let $\boldsymbol{\theta}^*$ denote a true value for $\boldsymbol{\theta}$. As $n \to \infty$, $\hat{\boldsymbol{\theta}} \xrightarrow{\text{p}} \boldsymbol{\theta}^*$.

That is for any $\epsilon > 0$

$$\mathbb{P}(|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*| > \epsilon) \to 0 \text{ as } n \to \infty.$$

↓

Convergence in probability

Intuition:

When you have enough sample data the estimator $\hat{\theta}$ is close to the population parameter θ^*.

MLE: asymptotic normality → HT CI

Let $\boldsymbol{\theta}^*$ denote a true value for $\boldsymbol{\theta}$.

Fisher information

$$\hat{\boldsymbol{\theta}} \approx^d N(\boldsymbol{\theta}^*, \mathcal{I}(\boldsymbol{\theta}^*)^{-1}).$$

approximately

MLE: asymptotic normality

The Fisher information is

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}\mathcal{J}(\boldsymbol{\theta}; \mathsf{Y}).$$

$\mathcal{I}(\boldsymbol{\theta})$ does not depend on y.

Observed infor.

The observed information is

Intuition: Information about the parameters in my data set.

$$\mathcal{J}(\boldsymbol{\theta}) = -\frac{\partial^2 l(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} = -\mathsf{H}_{l(\boldsymbol{\theta})},$$

where the hessian matrix, $\mathsf{H}_{l(\boldsymbol{\theta})}$, is a square matrix of second-order partial derivatives of the log likelihood.

For binomial regression with one predictor (e.g., $\boldsymbol{\theta} = \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$):

$$\mathcal{J}(\boldsymbol{\theta}) = \mathcal{J}(\boldsymbol{\beta}) = \begin{pmatrix} -\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_0^2} & -\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_1 \partial \beta_0} \\ -\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_0 \partial \beta_1} & -\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_1^2} \end{pmatrix}$$

Clearly $\mathcal{J}(\boldsymbol{\theta}) = \mathcal{J}(\boldsymbol{\theta}; \mathsf{y})$ depends on y through $l(\boldsymbol{\theta}) = l(\boldsymbol{\theta}; \mathsf{y})$.

MLE: asymptotic normality

The Fisher information is

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}\mathcal{J}(\boldsymbol{\theta}; \mathrm{Y}).$$

$\mathcal{I}(\boldsymbol{\theta})$ does not depend on y. → why?

We are averaging over all possible values of Y under the model we are considering.

Exercise: binomial regression with a logit link

We suppose that we observe $Y_i \sim \text{bin}(m_i, p_i)$, $i = 1, \ldots, n$, independent, where $p_i = \frac{1}{1+\exp(-\eta_i)}$ and $\eta_i = \beta_0 + \beta_1 x_i$. The log-likelihood is

$$\begin{aligned} l(\beta_0, \beta_1) &= c + \sum_i \left[y_i \log \frac{p_i}{1-p_i} + m_i \log(1-p_i) \right] \\ &= c + \sum_i \left[y_i(\beta_0 + \beta_1 x_i) - m_i \log(1 + e^{\beta_0 + \beta_1 x_i}) \right]. \end{aligned}$$

$p_i = \frac{1}{1+\exp(-\eta_i)}$ ← p_i is represented as the logistic function of the linear predictor η_i

Exercise: binomial regression with a logit link

↳ also called canonical link

Then,

$$\mathcal{J}(\beta) = \begin{pmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{pmatrix}.$$

See my derivation in "observed_information_binomial_regression.pdf".

So, since there are no y_i terms left,

$$\mathcal{I}(\beta) = \begin{pmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{pmatrix}.$$

↓

$\mathbb{E}(\mathcal{J}(\beta))$

Last week

- Binomial regression

$$Y \sim Bin(m, p)$$

$$g(p) = \eta = \beta_0 + \beta_1 t$$

$g(p)$: link function; $\beta_0 + \beta_1 t$: linear predictor

- MLE $\hat{\beta}_0, \hat{\beta}_1$
- Properties of MLE — asymptotic consistency, asymptotic normality
- Fisher information, observed fisher information

Today

- Continue Binomial regression
- Learn about CI
- Hypothesis testing

Observed information of binomial regression with a logit link

For a binomial regression with a logit link we have $y_i \sim \text{bin}(m_i, p_i)$, where $p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$.
The log-likelihood is given by

$$\begin{aligned} l(\beta_0, \beta_1) &= C + \sum_i \left[y_i \log \frac{p_i}{1-p_i} + m_i \log(1-p_i) \right] \\ &= C + \sum_i \left[y_i(\beta_0 + \beta_1 x_i) - m_i \log(1 + e^{\beta_0+\beta_1 x_i}) \right] \end{aligned}$$

log-likelihood

since

parts that do not contain β_0, β_1

$$\begin{aligned} \log \frac{p_i}{1-p_i} &= \beta_0 + \beta_1 x_i = \eta_i \\ 1 - p_i &= 1 - \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \\ &= \frac{1}{1 + e^{\beta_0+\beta_1 x_i}} \\ \log(1-p_i) &= -\log(1 + e^{\beta_0+\beta_1 x_i}). \end{aligned}$$

Now we compute the derivative.

$$\begin{aligned} \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} &= \sum_i \left[y_i - m_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_0^2} &= \sum_i \left[-m_i \frac{e^{\beta_0+\beta_1 x_i}(1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \frac{1}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ &= -\sum_i m_i p_i (1-p_i) \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_1 \partial \beta_0} &= \sum_i \left[-m_i \frac{e^{\beta_0+\beta_1 x_i} x_i (1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i} x_i}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i x_i \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i m_i x_i p_i (1-p_i) \\ \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} &= \sum_i \left[y_i x_i - m_i x_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_1^2} &= -\sum_i \left[m_i x_i \frac{e^{\beta_0+\beta_1 x_i} x_i (1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i} x_i}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i x_i^2 \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i m_i x_i^2 p_i (1-p_i) \end{aligned}$$

Note that

$$\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1\partial\beta_0} = \frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0\partial\beta_1}$$

Hence, the observed information is given by

$$\begin{aligned} J(\beta_0,\beta_1) &= \begin{bmatrix} -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0^2} & -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1\partial\beta_0} \\ -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0\partial\beta_1} & -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1^2} \end{bmatrix} \\ &= \begin{bmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{bmatrix} \end{aligned}$$

Please do these calculations!

Exercise: binomial regression with a logit link

Then,

$$\mathcal{J}(\boldsymbol{\beta}) = \begin{pmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{pmatrix}.$$

See my derivation in "observed_information_binomial_regression.pdf".

So, since there are no y_i terms left,

$$\mathcal{I}(\boldsymbol{\beta}) = \begin{pmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{pmatrix}.$$

$\mathbb{E}\left(\mathcal{J}(\boldsymbol{\beta})\right)$

MLE: asymptotic normality

Let $\boldsymbol{\theta}^*$ denote a true value for $\boldsymbol{\theta}$.

As $n \to \infty$,

$$\mathcal{I}(\boldsymbol{\theta}^*)^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \xrightarrow{\mathrm{d}} N(0, I).$$

That is,

$$\hat{\boldsymbol{\theta}} \approx^d N(\boldsymbol{\theta}^*, \mathcal{I}(\boldsymbol{\theta}^*)^{-1}).$$

second partial derivatives → curvature!

$$\mathcal{J}(\boldsymbol{\theta}) = -\frac{\partial^2 l(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} = -\mathrm{H}_{l(\boldsymbol{\theta})},$$

$\mathcal{I}(\theta) = \mathbb{E}[\mathcal{J}(\theta)]$

Any intuition for why variance is the inverse of fisher information which is related to hessian matrix?

The larger the $\mathcal{I}(\theta)$ is the smaller the variance!

larger curvature → sharper likelihood



MLE: asymptotic efficiency

$P(|\hat{\theta} - \theta^*| > \varepsilon) \xrightarrow{n \to \infty} 0$

- Asymptotic consistency: $\hat{\theta} \xrightarrow{\text{p}} \theta^*$ ->
- Asymptotic normality: $\hat{\theta} \approx^d N(\theta^*, \mathcal{I}(\theta^*)^{-1})$

MLE: asymptotically unbiased estimator with smallest variance $\mathcal{I}(\theta^*)^{-1}$.

In our Challenger example:

$\theta = \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ Linear combination $t = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$[0,1] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$

$\Rightarrow t^t \times \beta = \beta_1$

linear combination

Wald CI for $t^T\theta$ (linear combination of parameters)

$$t^T\theta = t_1\theta_1 + t_2\theta_2 + \cdots + t_m\theta_m$$

$$\boldsymbol{t} = \begin{pmatrix} t_1 \\ t_2 \\ \vdots \\ t_m \end{pmatrix} \quad \boldsymbol{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_m \end{pmatrix} \quad \boldsymbol{\theta}^* = \begin{pmatrix} \theta_1^* \\ \theta_2^* \\ \vdots \\ \theta_m^* \end{pmatrix} \quad \hat{\boldsymbol{\theta}} = \begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \vdots \\ \hat{\theta}_m \end{pmatrix}$$

$$\hat{\boldsymbol{\theta}} \approx N(\boldsymbol{\theta}^*, \mathcal{I}(\boldsymbol{\theta}^*)^{-1}).$$

In practice, we do not know $\boldsymbol{\theta}^*$, the true value of $\boldsymbol{\theta}$. Thus, we approximate $\mathcal{I}(\boldsymbol{\theta}^*)^{-1}$ using $\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}$, leading to

$$\hat{\boldsymbol{\theta}} \approx N(\boldsymbol{\theta}^*, \mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}).$$

$\hat{\boldsymbol{\theta}}$ → Our estimator

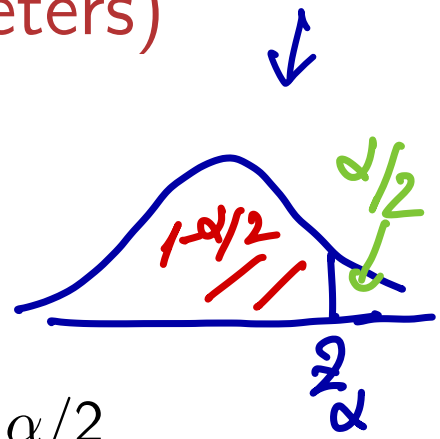
Then,

$$\boldsymbol{t}^T\hat{\boldsymbol{\theta}} \approx N(\boldsymbol{t}^T\boldsymbol{\theta}^*, \boldsymbol{t}^T\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}\boldsymbol{t}).$$

↳ linear combination of the parameters

Wald CI for $\boldsymbol{t}^T\boldsymbol{\theta}$ (linear combination of parameters)

$$\boldsymbol{t}^T\hat{\boldsymbol{\theta}} \approx N(\boldsymbol{t}^T\boldsymbol{\theta}^*, \boldsymbol{t}^T\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}\boldsymbol{t}).$$



An approximate $100(1-\alpha)\%$ confidence interval for $\boldsymbol{t}^T\boldsymbol{\theta}$:

$$\text{CI} = \boldsymbol{t}^T\hat{\boldsymbol{\theta}} \pm z_\alpha\sqrt{\boldsymbol{t}^T\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}\boldsymbol{t}}, \quad \text{where } \Phi(z_\alpha) = 1-\alpha/2$$

$1-\alpha$

$z_\alpha = z_{1-\alpha/2}$

In particular, taking $\boldsymbol{t}$ = standard unit vectors, $\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}, \begin{pmatrix}0\\1\\\vdots\\0\end{pmatrix}, \ldots, \begin{pmatrix}0\\0\\\vdots\\1\end{pmatrix}$,

we can obtain CI for each of parameters $\theta_1, \theta_2, \ldots, \theta_m$

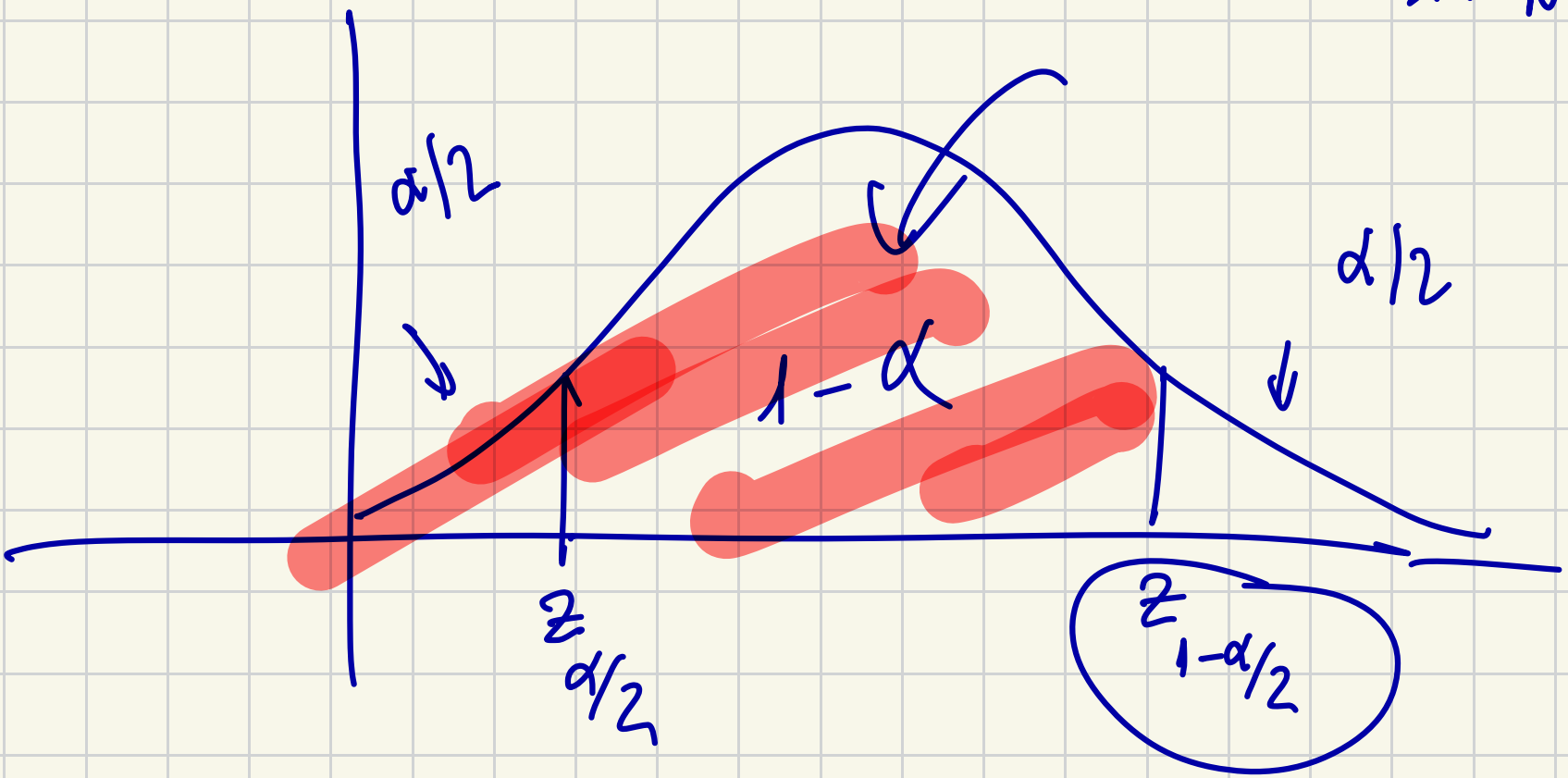
An approximate $100(1-\alpha)\%$ confidence interval for θ_i:

$$\hat{\theta}_i \pm z_\alpha\sqrt{(\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1})_{i,i}}$$



If $\mathcal{I}$ is unavailable then we can approximate it using the observed information $\mathcal{J}$, i.e., $\mathcal{I}(\hat{\boldsymbol{\theta}})^{-1} \approx \mathcal{J}(\hat{\boldsymbol{\theta}}; y)^{-1}$.

$X \sim N(0,1)$



$$1-\alpha/2 = P(X \leq z_{1-\alpha/2})$$

Reminder: questions from Challenger disaster

- Forecast probability of an O-ring being damaged when the launch temperature is 29 ^oF.
- How good is our forecast? Can we provide a confidence interval?
- Is temperature useful to predict the O-ring failing?

How good is our forecast? Can we provide a confidence interval?

$Y_i \sim Bin(m_i, p_i) \Rightarrow g(p_i) = \eta_i = \beta_0 + \beta_1 t_i$

logistic function

$\log\left(\frac{p_i}{1-p_i}\right) \Rightarrow p_i = \frac{e^{\eta_i}}{1+e^{\eta_i}}$

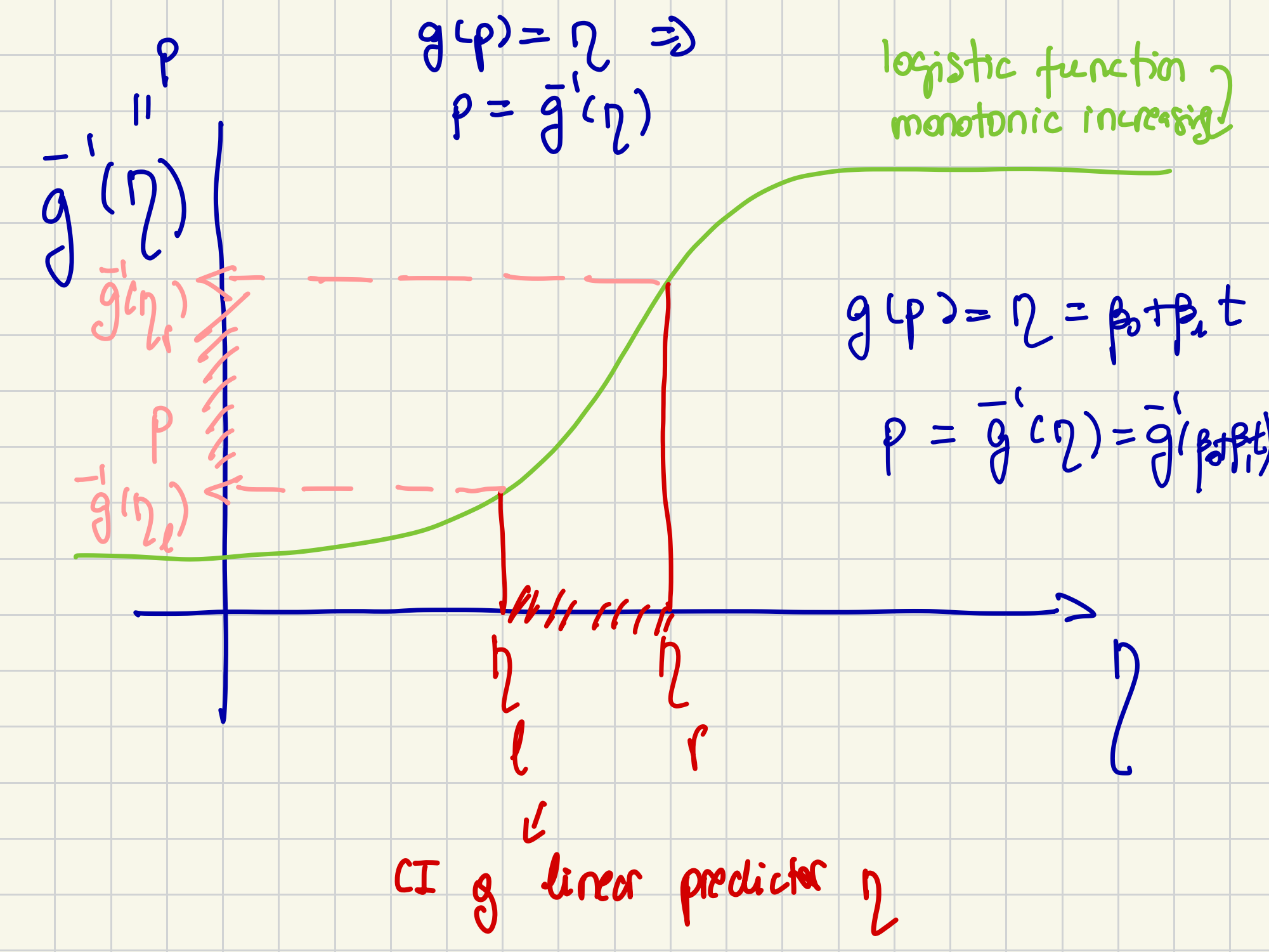
CI for $p = g^{-1}(\eta) = \frac{\exp(\eta)}{1+\exp(\eta)}$, where $\eta = \beta_0 + \beta_1 29$.

$= [1, 29] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$

- Step 1: Compute a CI for η, (η_l, η_r).
- Step 2: CI for $p = g^{-1}(\eta)$ is $(g^{-1}(\eta_l), g^{-1}(\eta_r))$.

CI for p: (0.864307, 0.9998686). See R script and result in "Confidence Interval for p" of Challenger.pdf

p → is not a linear combination of our parameters β_0 & β_1!


$g(p) = \eta \Rightarrow$
$p = g^{-1}(\eta)$
logistic function
monotonic increasing
p
$=$
$g^{-1}(\eta)$
$g^{-1}(\eta_r)$
p
$g^{-1}(\eta_l)$
$g(p) = \eta = \beta_0 + \beta_1 t$
$p = g^{-1}(\eta) = g^{-1}(\beta_0 + \beta_1 t)$
η_l
η_r
η
l
r
CI
g
linear predictor
η

```
##
##     Null deviance: 38.898  on 22  degrees of freedom
## Residual deviance: 16.912  on 21  degrees of freedom
## AIC: 33.675
##
## Number of Fisher Scoring iterations: 6
```

```
predict(logitmod, newdata=list(temp=29), type="response")
```

```
##         1
## 0.9954687
```

Confidence Interval for p

Compute standard errors

```
phat <- ilogit(betahat[1] + orings$temp*betahat[2])
I11 <- sum(6*phat*(1 - phat))
I12 <- sum(6*orings$temp*phat*(1 - phat))
I22 <- sum(6*orings$temp^2*phat*(1 - phat))
Iinv <- solve(matrix(c(I11, I12, I12, I22), 2, 2))
sqrt(Iinv[1,1])
```

```
## [1] 3.296634
```

```
sqrt(Iinv[2,2])
```

```
## [1] 0.05318407
```

Compute CI for eta = beta0 + beta1 29

```
si2 <- matrix(c(1, 29), 1, 2) %*% Iinv %*% matrix(c(1, 29), 2, 1)
etahat = betahat[1] + betahat[2]*29
eta_l = etahat - 2*sqrt(si2)
eta_r = etahat + 2*sqrt(si2)
etahat
```

```
## [1] 5.394493
```

```
c(eta_l, eta_r)
```

```
## [1] 1.851533 8.937452
```

Compute CI for p

```
ilogit (etahat)
```

```
## [1] 0.995479
```

```
c(ilogit (eta_l), ilogit (eta_r))
```

```
## [1] 0.8643070 0.9998686
```

Wald Test

Compute MLE

```
library(faraway)
data(orings)
logL <- function(beta, orings) {
```

log likelihood ratio CI

We have

$$2l(\hat{\boldsymbol{\theta}}) - 2l(\boldsymbol{\theta}^*) \approx \chi^2_k$$

where k is the dimension of $\boldsymbol{\theta}^*$.

This result can also, in principle, be used to construct a $100(1-\alpha)\%$ confidence region for $\boldsymbol{\theta}$:

$$\{\boldsymbol{\theta} : 2l(\hat{\boldsymbol{\theta}}) - 2l(\boldsymbol{\theta}) \leq \chi^2_k(1-\alpha)\}$$

where $\chi^2_k(1-\alpha)$ is the $100(1-\alpha)\%$ point for a χ^2_k distribution.

One lab problem will be to plot CI using the log likelihood ratio.

This approximation is generally better than the normal approximation for $\hat{\boldsymbol{\theta}}$. That is, it holds for smaller sample sizes.

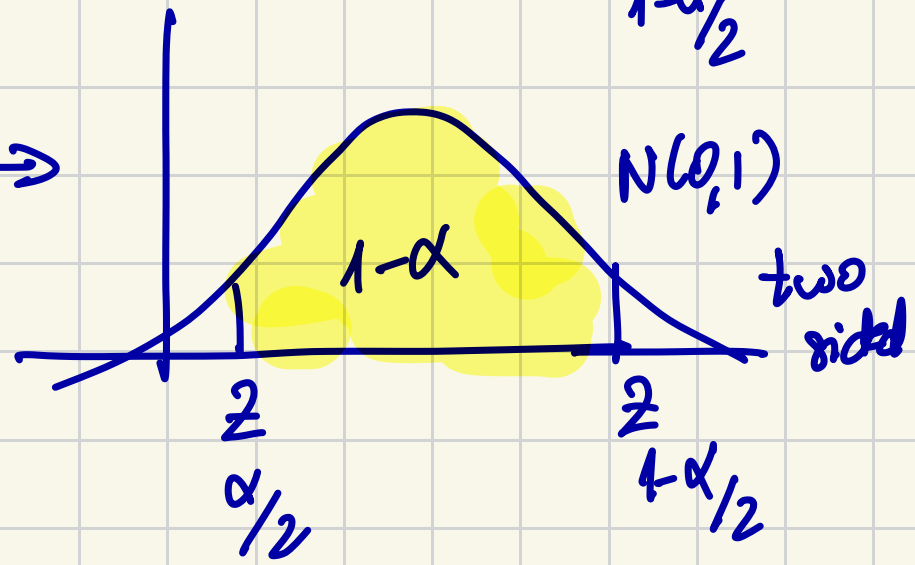
Intuition

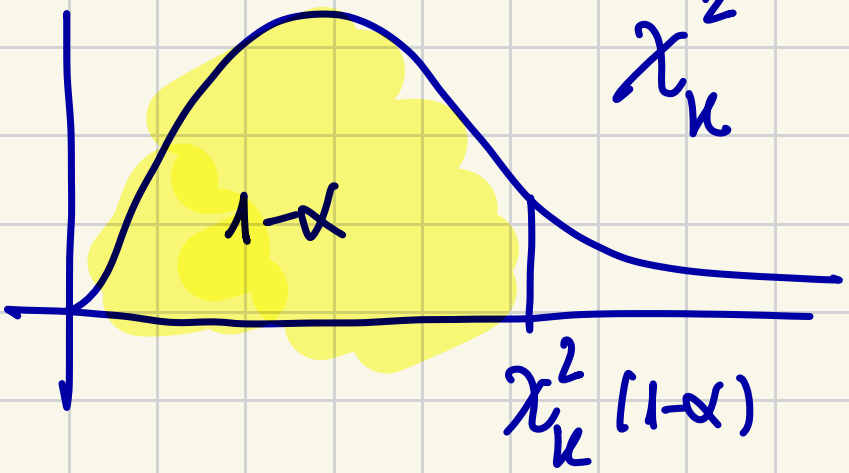
$\hat{\theta} \sim N(\theta, se^2(\hat{\theta})) \Rightarrow CI_{1-\alpha} = \hat{\theta} \pm z_{\alpha} se(\hat{\theta})$

$= z_{1-\alpha/2}$

$\Downarrow$

$$\frac{\hat{\theta} - \theta}{se(\theta)} \sim N(0,1) \rightarrow$$

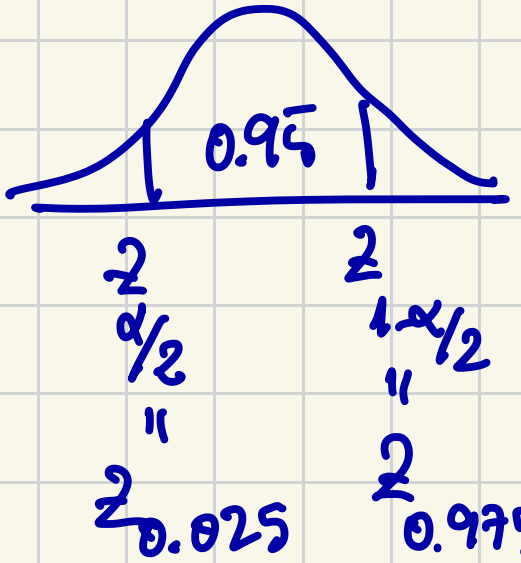




$$\{\theta : 2l(\hat{\theta}) - 2l(\theta) \leq \chi^2_k(1-\alpha)\}$$

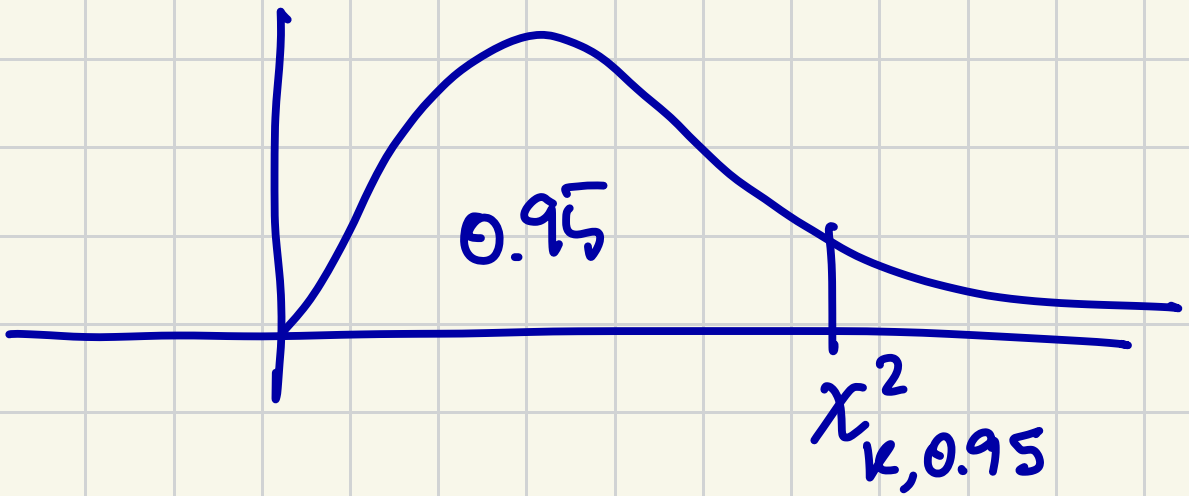
one sided

① $$\left| \frac{\theta - \hat{\theta}}{se(\hat{\theta})} \right| \leq 1.96$$



$$\boxed{\hat{\theta} \pm 1.96\, se(\hat{\theta})}$$

② $$\{\theta / 2\{l(\hat{\theta}) - l(\theta)\} \leq \chi^2_{k,0.95}\}$$



MLE: regularity conditions

l = log-likelihood

For maximum likelihood theory to hold we require

- l smooth enough with respect to $\boldsymbol{\theta}$ (third derivatives exist and continuous)
- Third order derivatives of l have bounded expectations
- Support of Y_i does not depend on $\boldsymbol{\theta}$
- The domain Θ of $\boldsymbol{\theta}$ is finite dimensional and doesn't depend on Y_i
- $\boldsymbol{\theta}^*$ is not on the boundary of Θ.

References

- McCullagh & Nelder (1989), Appendix A.
- F.W. Scholz, Maximum likelihood estimation. *Encyclopedia of Statistical Sciences* Vol. 7, p.4629ff. Wiley, 2006.